claim : $P'_{2n}(0) = 0$

proof : $t(1-t^2)^{-3/2} = \sum_{n=0}^{\infty} P_n'(0)\, t^n$

$$(1-t^2)^{-3/2} = \sum_{n=0}^{\infty} \binom{-3/2}{n} (1)^{-3/2-n} t^{2n}$$

$$= \sum_{n=0}^{\infty} \binom{-3/2}{n} t^{2n}$$

$$\binom{-3/2}{n} = \frac{\Gamma(-3/2+1)}{n!\,\Gamma(-3/2+1-n)} = \frac{\Gamma(-1/2)}{n!\,\Gamma(-1/2-n)}$$

$$= \frac{\Gamma(1/2)}{(-1/2)} \frac{1}{n!} \frac{(-1/2-n)}{\Gamma(1/2-n)}$$

$$= \frac{(2n+1)}{} \binom{-1/2}{n} = \frac{(2n+1)}{(n!)^2} \frac{(-1)^n (2n)!}{2^{2n}}$$

$$= \frac{(-1)^n (2n+1)}{2^{2n}} \frac{(2n)!}{(n!)^2} \Bigg\} \binom{2n}{n}$$

claim: $$\sum_{k=0}^{n} (2k+1) P_k(x) = P'_{n+1}(x) + P'_n(x)$$

proof:

Since $P'_{n+1}(x) - P'_{n-1}(x) = (2n+1) P_n(x)$

$$\sum_{k=1}^{n} (2k+1) P_k(x) = \sum_{k=1}^{n} \left[P'_{k+1}(x) - P'_{k-1}(x) \right]$$

$$= \sum_{k=0}^{n-1} \left[P'_{k+2}(x) - P'_k(x) \right] \qquad k \longrightarrow k+1$$

$P'_2 \quad P'_3 \quad P'_4 \quad \cdots \quad P'_{n+1}$ $\quad - \quad$ $P'_0 \quad P'_1 \quad P'_2 \quad \cdots \quad P'_{n-1}(x)$

$$= -P'_0(x) + P'_n(x) + P'_{n+1}(x)$$

$$\Rightarrow \underbrace{P'_0(x)}_{\text{0}} + \sum_{k=1}^{n} (2k+1) P_k(x) = P'_n(x) + P'_{n+1}(x)$$

$$P_0(x) = 1 = P_0'(x) + P_1'(x)$$

$$\Rightarrow \quad \sum_{k=0}^{n} (2k+1) P_k(k) = P_n'(x) + P_{n+1}'(x) .$$



Claim: $t^2(1-2xt+t^2)^{-3/2} = \sum_{n=0}^{\infty} [x P_n'(x) - n P_n(x)] t^n$

proof:

$$(1-2xt-t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) t^n$$

derivative w.r.t x.

$$(-2t)(-\tfrac{1}{2})(1-2xt-t^2)^{-3/2} = \sum_{n=0}^{\infty} P_n'(x) t^n$$

$$\Rightarrow \quad t(1-2xt-t^2)^{-3/2} = \sum_{n=0}^{\infty} P_n'(x) t^n$$

$$\Rightarrow \quad t^2(1-2xt-t^2)^{-3/2} = \sum_{n=0}^{\infty} P_n'(x) t^{n+1}$$

$n \to n-1$

$$= \sum_{n=1}^{\infty} P_{n-1}'(x) t^n$$

Since $x P_n'(x) - n P_n(x) = P_{n-1}'(x)$

$$= \sum_{n=1}^{\infty} [x P_n'(x) - n P_n(x)] t^n$$

Claim: $\int_{-1}^{1} P_n(x)\,dx = 0$

or using orthonormality relation $n \neq 0$

proof: $\alpha_n \int_{-1}^{1} (1) \frac{d^n}{dx^n}\left[(x^2-1)^n\right] dx$

$= \alpha_n \frac{d^{n-1}}{dx^{n-1}}\left[(x^2-1)^n\right]\Big|_{-1}^{1}$

$= \alpha_n \frac{d^{n-1}}{dx^{n-1}}\left(\sum_{k=0}^{n}\binom{n}{k} x^{2k}\right)\Big|_{-1}^{1}$

$= \sum_{k=0}^{n} \alpha_n \binom{n}{k} \frac{d^{n-1}}{dx^{n-1}}\left(x^{2k}\right)\Big|_{-1}^{1}$

if $2k < n-1 \Rightarrow 0$

if $2k > n-1$ $\quad d^{n-1}(x^{2k}) = (2k)(2k-1)\cdots(2k-n+1)\, x^{2k-n}$

n even

n - odd $\qquad 2k \geq n-1 \qquad k \geq \frac{n-1}{2}$

$$\Rightarrow \sum_{k=\left(\frac{n-1}{2}\right)}^{n} \alpha_n \binom{n}{k} (2k)\cdots(2k-n+1)\, x^{2k-n} \Big|_{-1}^{1}$$

n is odd

$$\frac{(-1)^n}{=} = \boxed{-1}$$

$$\Rightarrow \quad 0$$

claim: $\int_{-1}^{1} x P_n(x) P_{n-1}(x)\, dx = \frac{2n}{4n^2-1}$

proof: $= \frac{(n+1)}{(2n+1)} \int \cancel{P_{n+1}} P_{n-1}\, dx$ (0)

$+ \frac{n}{(2n+1)} \int_{-1}^{1} [P_{n-1}(x)]^2\, dx$

$= \frac{n}{2n+1} \times \frac{2}{(2(n-1)+1)} = \frac{2n}{(2n+1)(2n-1)}$

---

claim: $\int_{-1}^{1} P_n(x) P_{n+1}'(x)\, dx = 2$

$P_{n+1}'(x) = (n+1) P_n(x) + x P_n'(x)$

$= (n+1) P_n(x) + P_{n-1}'(x) + n P_n(x)$

$= (2n+1) P_n(x) + P_{n-1}'(x).$

$$\Rightarrow \quad 2 + \int_{-1}^{1} P_n(x)\, P_{n-1}'(x)\, dx$$

$$P_{n-1}'(x) = \underline{P_{n-2}'(x)} + (2n-3)\, P_{n-2}(x)$$

$$\Rightarrow \quad 2 + \int_{-1}^{1} P_n(x)\, P_{n-2}'(x)\, dx$$

$$\vdots$$

$$2 + \int_{-1}^{1} P_n(x) \underbrace{P_0'(x)}_{0}\, dx \quad = 2$$

---

claim : $\displaystyle\int_{-1}^{1} x\, P_n'(x) P_n(x)\, dx = \frac{2n}{2n+1}$

proof : $\quad x\, P_n'(x) = P_{n-1}'(x) + n\, P_n(x)$

$$\Rightarrow \quad n \int_{-1}^{1} P_n^2(x) = \frac{2n}{2n+1}$$

$$\int_{-1}^{1} (1-x^2)\, P_n'(x)\, P_k'(x)\, dx = 0$$

$$= \underbrace{(1-x^2)\, P_n'(x)\, P_k(x)\Big|_{-1}^{1}}_{0} - \int_{-1}^{1} \frac{d}{dx}\left[(1-x^2)\, P_n'(x)\right] P_k(x)\, dx$$

$$\frac{d}{dx}\left[(1-x^2)\, P_n'(x)\right] = n(n+1)\, P_n(x)$$

$$= -(n+1)(n) \int_{-1}^{1} P_n(x)\, P_k(x) = 0 \quad \text{for } k \neq n$$

---

claim:

$$\int_{-1}^{1} (1-2xt+t^2)^{-1/2} P_n(x)\, dx = \frac{2t^n}{2n+1}$$

$$(1-2xt+t^2)^{-1/2} = \sum_{k=0}^{\infty} P_k(x)\, t^k$$

$$= \int_{-1}^{1} \sum_{k=0}^{\infty} P_k(x)\, t^k\, P_n(x)\, dx$$

$$= \sum_{k=0}^{\infty} t^k \int_{-1}^{1} P_k(x)\, P_n(x)\, dx$$

$$= \frac{t^n (2)}{(2n+1)}$$

claim: $$\int_{-1}^{1} x^2 P_{n+1}(x) P_{n-1}(x)\, dx = \frac{2n(n+1)}{(4n^2-1)(2n+3)}$$

$n = 1, 2, \ldots$

proof: $$x P_n(x) = \frac{(n+1)}{(2n+1)} P_{n+1}(x) + \frac{n}{(2n+1)} P_{n-1}(x)$$

$n \longrightarrow n-1$

$$x P_{n-1}(x) = \frac{n}{(2n-1)} P_n(x) + \frac{(n-1)}{(2n-1)} P_{n-2}(x)$$

$$x P_{n+1}(x) = \frac{(n+2)}{(2n+3)} P_{n+2}(x) + \frac{n+1}{(2n+3)} P_n(x)$$

$$\Rightarrow \frac{(n+1)}{(2n+3)} x \frac{n}{(2n-1)} \int_{-1}^{1} P_n^2\, dx$$

$$= \frac{(n+1)}{(2n+3)} x \frac{n}{(2n-1)} x \frac{2}{(2n+1)}$$

Claim: $$\int_{-1}^{1} f(x) P_n(x)\,dx = \frac{(-1)^n}{2^n n!} \int_{-1}^{1} f^{(n)}(x)(x^2-1)^n dx$$

proof:

$$\frac{1}{2^n n!} \int_{-1}^{1} f(x) \frac{d^n}{dx^n}(x^2-1)^n$$

$$= \frac{1}{2^n n!} \left[\underbrace{f(x) \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n \Big|_{-1}^{1}}_{=0 \text{ why?!}} + (-1)\int_{-1}^{1} f^{(1)} \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n \right]$$

$$= \left[\frac{(-1)^1}{2^n n!} \int_{-1}^{1} f^{(1)} \frac{d^{n-1}}{dx^{n-1}}(x^2-1)^n \right]$$

$$= \frac{(-1)^n}{2^n n!} \int_{-1}^{1} f^{(n)} (x^2-1)^n$$